IMO 2026 — Autonomous Solutions

GPT-5.6 Sol Pro

xhigh reasoning effort

Independently graded: 37 / 42

P1 7 P2 7 P3 2 P4 7 P5 7 P6 7

All six problems attempted fully autonomously in one identical minimal agent loop (bash / write_file / read_file), network blocked, 150 min per-problem cap. Scores are IMO-style 0–7 from independent verifier agents that re-derived key algebra symbolically and machine-tested claims — not the model's self-reports. Per-turn audit trails, grader verdicts, and run statistics accompany this document in the `imo-2026` repository.

IMO 2026 — Problem 1

Solution by GPT-5.6 Sol Pro (xhigh reasoning)

2026-07-22

Source: `results/gpt-5.6-sol-pro/problem-01/current.md` — autonomous run, no human intervention; solution as written by the model.

Problem

There are 2026 integers greater than 1 written on a blackboard, not necessarily different. In a move, Confucius chooses two integers $m > 1$ and $n > 1$ from different places on the blackboard and replaces these two integers with $\gcd(m, n)$ and $\frac{\operatorname{lcm}(m,n)}{\gcd(m,n)}$. He continues to make moves while it is possible to do so. (a) Prove that, regardless of the choices of Confucius, after finitely many moves, exactly one integer M on the blackboard is greater than 1. (b) Prove that the value of M does not depend on the choices of Confucius.

Solution document

Status

solved

Problem

There are 2026 integers greater than 1 written on a blackboard, not necessarily different. In a move, Confucius chooses two integers $m > 1$ and $n > 1$ from different places on the blackboard and replaces these two integers with $\gcd(m, n)$ and $\frac{\operatorname{lcm}(m,n)}{\gcd(m,n)}$. He continues to make moves while it is possible to do so. (a) Prove that, regardless of the choices of Confucius, after finitely many moves, exactly one integer M on the blackboard is greater than 1. (b) Prove that the value of M does not depend on the choices of Confucius.

Approaches tried

- **Product-and-count descent (successful):** Track the product P of all entries and the number K of nonunit entries. A move with selected gcd g changes P to P/g. If $g > 1$, the product decreases; if $g = 1$, the product is fixed but K decreases. Thus $P + K$ strictly decreases.

- **Prime-adic Euclidean invariant (successful):** For each prime p, the selected exponents (a,b) become $(\min(a,b), |a-b|)$. The gcd of the complete list of p-adic exponents is invariant under this Euclidean-algorithm step, and at termination it is exactly $v_p(M)$.
- **Exhaustive sanity check:** A Python search over all boards of lengths $2,3,4$ with entries from 2 through 10 found that every possible play terminates at the value predicted by the prime-adic formula.

Current best

Let the initial entries be $A_1, \dots, A_{2026}$. Every play terminates with one nonunit, and its choice-independent value is

$$\boxed{M = \prod_p p^{\gcd(v_p(A_1),\dots,v_p(A_{2026}))}},$$

where $\gcd(0,\dots,0) = 0$ and the product is effectively only over primes dividing at least one initial entry. Termination follows from strict descent of $P+K$, while the formula follows from preservation, prime by prime, of the gcd of all valuation exponents.

Full proof

Let a **nonunit** mean an entry greater than 1. At any stage, define

$P =$ the product of all entries on the blackboard, $\qquad K =$ the number of nonunit entries.

Both P and K are positive integers as long as there is at least one nonunit.

Suppose that Confucius selects $m, n > 1$, and put $g = \gcd(m,n)$. Since

$$mn = \gcd(m,n)\operatorname{lcm}(m,n) = g\operatorname{lcm}(m,n),$$

the product of the two selected entries changes from mn to

$$g \cdot \frac{\operatorname{lcm}(m,n)}{g} = \operatorname{lcm}(m,n) = \frac{mn}{g}.$$

Consequently, the total product changes from P to P/g.

The number K cannot increase, since two nonunits are replaced by only two entries. More precisely:

- If $g > 1$, then $P/g < P$, while K does not increase.
- If $g = 1$, the two new entries are 1 and mn. Because $mn > 1$, the product P is unchanged and K decreases by exactly one.

It follows that the positive integer $P + K$ strictly decreases after every move. Hence there cannot be infinitely many moves.

Furthermore, no move can eliminate all nonunits. Indeed, if $g > 1$, then the first new entry g is a nonunit; if $g = 1$, then the second new entry mn is a nonunit. Since initially there are nonunits, there is therefore always at least one. Once the process stops, there cannot be two nonunits, since any two of them could be selected for another move. Thus at termination there is exactly one nonunit. Denote it by M. This proves part (a).

It remains to determine M. Write the initial entries as

$$A_1, A_2, \dots, A_{2026}.$$

Fix an arbitrary prime p. For any positive integer x, let $v_p(x)$ denote the exponent of p in its prime factorization. If the selected entries m, n have exponents

$$a = v_p(m), \qquad b = v_p(n),$$

then the two new entries have exponents

$$v_p(\gcd(m,n)) = \min(a,b)$$

and

$$v_p\left(\frac{\operatorname{lcm}(m,n)}{\gcd(m,n)}\right) = \max(a,b) - \min(a,b) = |a-b|.$$

The gcd of these two new exponents equals the gcd of the old exponents. Indeed, if, without loss of generality, $a \ge b$, then

$$\gcd(\min(a,b), |a-b|) = \gcd(b, a-b) = \gcd(a,b),$$

because an integer divides both a and b if and only if it divides both b and $a-b$.

Therefore the quantity

$$G_p = \gcd(v_p(x_1), v_p(x_2), \dots, v_p(x_{2026})),$$

where $x_1, \dots, x_{2026}$ are the current entries, is invariant under a move: the gcd of the two altered exponents is preserved, and all the other exponents are unchanged. We use the convention that the gcd of a list consisting entirely of zeros is 0.

At the end, the entries are M and 2025 copies of 1, in some order. Their p-adic exponents are consequently $v_p(M)$ and 2025 zeros. Their gcd is $v_p(M)$. Applying the invariant from the initial board to the final board gives

$$v_p(M) = \gcd(v_p(A_1), v_p(A_2), \dots, v_p(A_{2026})) \qquad (1)$$

for every prime p.

Only primes dividing the finite product $A_1 A_2 \cdots A_{2026}$ can occur in (1). Hence unique factorization yields the finite formula

$$M = \prod_p p^{\gcd(v_p(A_1), v_p(A_2), \dots, v_p(A_{2026}))}.$$

The right-hand side depends only on the initial entries, not on any choices of pairs during the process. Therefore the final value of M is independent of all of Confucius's choices, proving part (b). $\square$

IMO 2026 — Problem 2

Solution by GPT-5.6 Sol Pro (xhigh reasoning)

2026-07-23

Source: `results/gpt-5.6-sol-pro/problem-02/current.md` — autonomous run, no human intervention; solution as written by the model.

Problem

Let ABC be a triangle and let points M and N be the midpoints of sides AB and AC, respectively. Let points K and L be chosen inside triangles BMC and BNC, respectively, such that K lies inside the angle LBA, L lies inside the angle ACK, and $\angle KBA = \angle ACL$, $\angle LBK = \angle LNC$, $\angle LCK = \angle BMK$. Let O be the circumcentre of triangle AKL. Prove that $OM = ON$.

Solution document

Status

solved

Problem

Let ABC be a triangle and let points M and N be the midpoints of sides AB and AC, respectively. Let points K and L be chosen inside triangles BMC and BNC, respectively, such that K lies inside the angle LBA, L lies inside the angle ACK, and $\angle KBA = \angle ACL$, $\angle LBK = \angle LNC$, $\angle LCK = \angle BMK$. Let O be the circumcentre of triangle AKL. Prove that $OM = ON$.

Approaches tried

- **Angle parameters and polar coordinates.** The midpoint conditions and sine rule reduce the problem to a trigonometric factorization.
- **Circle intersections with AB, AC.** If the circumcircle of AKL meets the oriented lines AB, AC again at D, E, then $OM = ON$ is exactly $AB \cdot AD - AC \cdot AE = (AB^2 - AC^2)/2$.
- **Numerical/symbolic checking.** `code/explore_trig.py` checked compatible samples; `code/symbolic_poly.py` exposed the exact harmonic factorization and its factor $\sin(x - y)$.
- **Harmonic factorization (successful).** The target residual is exactly $\sin(x - y)$ times the compatibility residual. A complete proof is in `lemmas/trigonometric-factorization.md`.

Current best

The problem is solved. With suitable angle parameters, the two midpoint conditions determine AK/AB and AL/AC. The two remaining angle conditions yield a compatibility equation. An oriented-circle calculation turns $OM = ON$ into a trigonometric target, and the factorization lemma shows that its residual is $\sin(x - y)$ times the already-zero compatibility residual.

Full proof

Let

$$p = \angle KBA = \angle ACL, \quad q = \angle LBK = \angle LNC, \quad r = \angle LCK = \angle BMK,$$

and put

$$x = \angle BAK, \qquad y = \angle CAL, \qquad \delta = \angle KAL.$$

Because K lies inside $\angle LBA$, the ray BK lies between BA and BL; because L lies inside $\angle ACK$, the ray CL lies between CA and CK. We first justify the resulting order of the rays at A. Write the positive normalized affine coordinates of K, L relative to A, B, C as

$$K = (k_A, k_B, k_C), \qquad L = (l_A, l_B, l_C).$$

At B, the first betweenness says $k_C/k_A < l_C/l_A$; at C, the second says $l_B/l_A < k_B/k_A$. Therefore

$$\frac{k_B}{l_B} > \frac{k_A}{l_A} > \frac{k_C}{l_C}.$$

Thus $k_C/k_B < l_C/l_B$. Indeed, a ray from A through a point with affine coordinates (u_A, u_B, u_C) has direction $u_B\overrightarrow{AB} + u_C\overrightarrow{AC}$; consequently this last inequality says precisely that the ray AK lies between AB and AL. Since both points are inside ABC, the ray order is consequently AB, AK, AL, AC. Hence

$$\angle BAC = x + \delta + y. \qquad (0)$$

Put $c = AB$ and $b = AC$.

Applying the sine rule in ABK and BMK, and using $BM = c/2$, gives

$$BK = \frac{c\sin x}{\sin(p + x)} = \frac{c\sin r}{2\sin(p + r)}.$$

Consequently

$$\cot x = \cot p + 2\cot r, \qquad AK = cP, \qquad P := \frac{\sin p}{\sin(p + x)}. \qquad (1)$$

Similarly, the sine rule in ACL and CNL, using $CN = b/2$, gives

$$\cot y = \cot p + 2\cot q, \qquad AL = bQ, \qquad Q := \frac{\sin p}{\sin(p + y)}. \qquad (2)$$

Now apply the sine rule in ACK and ABL. From (0),

$$\angle CAK = \delta + y, \qquad \angle BAL = x + \delta.$$

Also, the angle assumptions give

$$\angle ACK = p + r, \qquad \angle ABL = p + q.$$

Thus, on setting

$$U = p + r + \delta + y, \qquad V = p + q + \delta + x,$$

we obtain

$$\frac{c}{b}P = \frac{\sin(p+r)}{\sin U}, \qquad \frac{b}{c}Q = \frac{\sin(p+q)}{\sin V}. \qquad (3)$$

Multiplying these equalities yields

$$PQ \sin U \sin V = \sin(p+r)\sin(p+q). \qquad (4)$$

Let the circle through A, K, L meet the oriented lines AB, AC again at D, E, respectively, and write $AD = d$, $AE = e$ as directed lengths. Take A as origin, and let $\mathbf{e}_B, \mathbf{e}_C$ be the unit vectors along AB, AC. If $\mathbf{w} = 2\overrightarrow{AO}$, the circle has equation

$$|\mathbf{z}|^2 = \mathbf{w} \cdot \mathbf{z}.$$

It follows that

$$d = \mathbf{w} \cdot \mathbf{e}_B, \qquad e = \mathbf{w} \cdot \mathbf{e}_C. \qquad (5)$$

Since $\overrightarrow{AM} = (c/2)\mathbf{e}_B$ and $\overrightarrow{AN} = (b/2)\mathbf{e}_C$, we have

$$OM^2 - ON^2 = \frac{c^2 - b^2}{4} - \frac{cd - be}{2}.$$

Therefore $OM = ON$ is equivalent to

$$cd - be = \frac{c^2 - b^2}{2}. \qquad (6)$$

Let $\mathbf{u}_K, \mathbf{u}_L$ be the unit vectors along AK, AL. The circle equation at K, L says

$$\mathbf{w} \cdot \mathbf{u}_K = cP, \qquad \mathbf{w} \cdot \mathbf{u}_L = bQ.$$

The angles from AB to AK, from AK to AL, and from AL to AC are x, δ, y, respectively. Solving these two linear equations and then taking the projections of $\mathbf{w}$ on AB, AC gives

$$d = \frac{cP\sin(\delta + x) - bQ\sin x}{\sin\delta}, \qquad e = \frac{-cP\sin y + bQ\sin(\delta + y)}{\sin\delta}. \qquad (7)$$

From the first relation in (1), a direct sine expansion gives

$$P\sin(\delta + x) - \frac{1}{2}\sin\delta = \frac{\sin p}{2\sin(p+r)}\sin(\delta + r). \qquad (8)$$

Analogously, (2) gives

$$Q\sin(\delta + y) - \frac{1}{2}\sin\delta = \frac{\sin p}{2\sin(p+q)}\sin(\delta + q). \qquad (9)$$

Substitution of (7)–(9) shows that (6) is equivalent to

$$\frac{c}{b}\frac{\sin p\sin(\delta+r)}{2\sin(p+r)}-\frac{b}{c}\frac{\sin p\sin(\delta+q)}{2\sin(p+q)}+P\sin y-Q\sin x=0. \qquad (10)$$

Using (3), equation (10) is equivalent to $T=0$, where

$$\begin{aligned}T&=Q\sin p\sin(\delta+r)\sin V-P\sin p\sin(\delta+q)\sin U\\&\quad+2(P\sin y-Q\sin x)\sin(p+r)\sin(p+q). \qquad (11)\end{aligned}$$

We use the following trigonometric identity, proved below:

$$T=\sin(x-y)\big(PQ\sin U\sin V-\sin(p+r)\sin(p+q)\big). \qquad (12)$$

Indeed, the relations $\cot x=\cot p+2\cot r$ and $\cot y=\cot p+2\cot q$ imply

$$P=\frac{\sin r}{2\sin(r-x)}, \qquad Q=\frac{\sin q}{2\sin(q-y)}, \qquad (13)$$

and

$$\frac{\sin(p+r)}{\sin p}=\frac{\sin(r-x)}{\sin x}, \qquad \frac{\sin(p+q)}{\sin p}=\frac{\sin(q-y)}{\sin y}. \qquad (14)$$

From (13), putting the expression in brackets below over a common denominator, and using the sine subtraction formula, we obtain

$$\begin{aligned}&[2(P\sin y-Q\sin x)+\sin(x-y)]\sin(p+r)\sin(p+q)\\&\quad=\sin^2 p\sin(q+x-r-y). \qquad (15)\end{aligned}$$

For completeness, the numerator used here is

$$\begin{aligned}&\sin r\sin y\sin(q-y)-\sin q\sin x\sin(r-x)\\&\quad+\sin(x-y)\sin(r-x)\sin(q-y)\\&=\sin x\sin y\sin(q+x-r-y),\end{aligned}$$

and (14) then gives (15).

It remains to verify (12). Put

$$W=2\delta+p+q+r, \qquad D=q+x-r-y.$$

After applying $2\sin s\sin t=\cos(s-t)-\cos(s+t)$, the part of twice the difference between the two sides of (12) that depends on δ is

$$-Q\sin p\cos(W+x)+P\sin p\cos(W+y)+\sin(x-y)PQ\cos(W+p+x+y). \qquad (16)$$

Divide (16) by PQ and use $\sin p/P=\sin(p+x)$ and $\sin p/Q=\sin(p+y)$. The result is

$$-\sin(p+x)\cos(W+x)+\sin(p+y)\cos(W+y)+\sin(x-y)\cos(W+p+x+y),$$

which is zero: applying $2\sin a\cos b=\sin(a+b)+\sin(a-b)$ makes its six terms cancel in pairs.

It remains to compare the constant parts. We first prove

$$Q \sin p \cos(r - p - q - x) - P \sin p \cos(q - p - r - y)$$
$$- \sin(x - y) PQ \cos D = -2 \sin^2 p \sin D. \qquad (17)$$

After division by PQ, this is equivalent to

$$\sin(p + x) \cos(p + y + D) - \sin(p + y) \cos(D - p - x)$$
$$- \sin(x - y) \cos D = -2 \sin(p + x) \sin(p + y) \sin D. \qquad (18)$$

Here we used $r - p - q - x = -(p + y + D)$ and $q - p - r - y = D - (p + x)$. To prove (18), expand the two cosines of sums and also $\sin(x - y) = \sin((p + x) - (p + y))$. All terms containing $\cos D$ cancel, while the two remaining terms are both $-\sin(p + x) \sin(p + y) \sin D$.

In twice the difference between the sides of (12), the complete constant part is the left side of (17) plus

$$[4(P \sin y - Q \sin x) + 2 \sin(x - y)] \sin(p + r) \sin(p + q).$$

By (15), this latter expression is $2 \sin^2 p \sin D$, which cancels the right side of (17). Thus both the variable and constant parts vanish, proving (12).

By (4), the right-hand side of (12) is zero. Hence $T = 0$, so (10), (6), and therefore $OM = ON$ follow.

IMO 2026 — Problem 3

Solution by GPT-5.6 Sol Pro (xhigh reasoning)

2026-07-22

Source: `results/gpt-5.6-sol-pro/problem-03/current.md` — autonomous run, no human intervention; solution as written by the model.

Problem

Let n be a positive integer. Liu Bang and Xiang Yu have a stick of length 1 and want to divide it between themselves. Liu Bang marks at most n points on the stick, and then Xiang Yu marks at most n points on the stick. The marked points are distinct. Then, the stick is cut at all marked points, creating a number of pieces. Afterwards, they take turns claiming any unclaimed piece of the stick, with Liu Bang going first. Each player's goal is to maximise the total length of their own pieces. For each n, determine the largest value c such that Liu Bang may guarantee a total length of at least c, regardless of Xiang Yu's play.

Solution document

Status

partial

Problem

Let n be a positive integer. Liu Bang and Xiang Yu have a stick of length 1 and want to divide it between themselves. Liu Bang marks at most n points on the stick, and then Xiang Yu marks at most n points on the stick. The marked points are distinct. Then, the stick is cut at all marked points, creating a number of pieces. Afterwards, they take turns claiming any unclaimed piece of the stick, with Liu Bang going first. Each player's goal is to maximise the total length of their own pieces. For each n, determine the largest value c such that Liu Bang may guarantee a total length of at least c, regardless of Xiang Yu's play.

Approaches tried

- **Ordered-draft reduction (proved):** for fixed final lengths $x_1 \geq \cdots \geq x_m$, largest-remaining-piece strategies give Liu the game value $x_1 + x_3 + \cdots = \frac{1}{2}(1 + D)$, where $D = x_1 - x_2 + x_3 - \cdots$. A complete proof is in `lemmas/draft-value.md`.

- **Layer-cake/parity formulation (active):** $D = \int_0^\infty (N(t) \bmod 2)\, dt$, where $N(t)$ is the number of pieces of length at least t.
- **Pairing/graph formulation (exploratory):** if all but one final pieces can be paired by equal length, then D is the singleton length. Xiang's refinements can be represented as distributions of paired weights among Liu's initial intervals.
- **Integer-grid brute force (conjecture found):** dependency-free exhaustive searches for $n = 2, 3, 4$ point to Liu's optimal initial lengths being proportional to $1, 2, 4, \dots, 2^n$. The grid output is stored under `scratch/grid_n*.txt`; grid effects make the computed excess slightly exceed the continuous limit.
- **Small case $n = 1$ (proved):** the optimal initial lengths are $1/3, 2/3$, and $c_1 = 2/3$.

Current best

The conjectured explicit answer is

$$\boxed{c_n = \frac{2^n}{2^{n+1} - 1}}.$$

Equivalently, the conjectured optimal guaranteed alternating excess is $D = 1/(2^{n+1} - 1)$. Liu should cut the stick into lengths $1, 2, 4, \dots, 2^n$, normalized by their sum $2^{n+1} - 1$. The remaining open gap is a complete proof of the partition-refinement theorem: among all multisets of at most $n + 1$ initial lengths totaling 1, the maximum possible minimum alternating excess after at most n splits is $1/(2^{n+1} - 1)$.

Full proof

The complete general proof is not yet available. The established reduction follows.

Let the final lengths be $x_1 \geq \cdots \geq x_m$. If Liu always takes a largest remaining piece, then before his jth move at most $2j - 2$ pieces have disappeared, so he receives at least x_{2j-1}. Conversely, if Xiang always takes a largest remaining piece, then on his jth move he receives at least x_{2j}, leaving Liu at most $\sum_j x_{2j-1}$. Hence the value of the claiming phase is

$$F = x_1 + x_3 + \cdots = \frac{1 + D}{2}, \qquad D = x_1 - x_2 + x_3 - x_4 + \cdots.$$

Writing $N(t) = \#\{i : x_i \geq t\}$ and integrating horizontal layers gives

$$D = \int_0^\infty (N(t) \bmod 2)\, dt.$$

It remains to prove the stated extremal refinement theorem.

For $n = 1$, let Liu's two initial pieces have lengths $a \geq b = 1 - a$. After Xiang cuts one piece, there are three pieces and Liu receives the largest plus the smallest, i.e. 1 minus the middle piece. If $a \leq 2/3$, Xiang can cut the a-piece into lengths arbitrarily close to $a - b$ and b, so the middle length can approach b and Liu's infimum payoff is $1 - b = a$. If $a \geq 2/3$, no cut can create two pieces both exceeding $a/2$, while cutting the a-piece nearly in half makes the middle length approach $a/2$; hence Liu's infimum payoff is $1 - a/2$. Their maximum is $2/3$ at $a = 2/3$.

IMO 2026 — Problem 4

Solution by GPT-5.6 Sol Pro (xhigh reasoning)

2026-07-22

Source: `results/gpt-5.6-sol-pro/problem-04/current.md` — autonomous run, no human intervention; solution as written by the model.

Problem

Shan-Yu and Mulan are playing a game. Let θ be an angle with $0^\circ < \theta < 180^\circ$ known to both players. Initially, Shan-Yu makes a paper triangle $\mathcal{T}$ with measurements of his choice. Then, they repeatedly perform the following steps: If $\mathcal{T}$ has at least one angle measuring exactly θ, then the game stops and Mulan wins. Otherwise, Mulan chooses a point P on the perimeter of $\mathcal{T}$, different from its three vertices. She then makes a straight cut from P to the opposite vertex of $\mathcal{T}$, splitting it into two triangles. Shan-Yu discards one of the two triangles. The remaining triangle becomes the new $\mathcal{T}$. For which real values of θ can Mulan guarantee her victory in finitely many steps, no matter how Shan-Yu plays?

Solution document

Status

solved

Problem

Shan-Yu and Mulan are playing a game. Let θ be an angle with $0^\circ < \theta < 180^\circ$ known to both players. Initially, Shan-Yu makes a paper triangle $\mathcal{T}$ with measurements of his choice. Then, they repeatedly perform the following steps: If $\mathcal{T}$ has at least one angle measuring exactly θ, then the game stops and Mulan wins. Otherwise, Mulan chooses a point P on the perimeter of $\mathcal{T}$, different from its three vertices. She then makes a straight cut from P to the opposite vertex of $\mathcal{T}$, splitting it into two triangles. Shan-Yu discards one of the two triangles. The remaining triangle becomes the new $\mathcal{T}$. For which real values of θ can Mulan guarantee her victory in finitely many steps, no matter how Shan-Yu plays?

Approaches tried

- Modeled a triangle by its three angle measures in units of θ. A cut from a vertex of angle a splitting it into x and $a-x$ produces angle triples $(b, x, s-b-x)$ and $(c, a-x, b+x)$, where

$s = 180°/\theta$.
- First observed that $90°$ is universally constructible by an altitude, and hence that $90°/n$ works. This was useful but not the full family: for instance $60°$ can be forced by making the two supplementary angles at the cut point integral multiples of $60°$.
- Developed the integral-angle strategy. If $s = N$ is an integer, any positive integral multiple of θ is a winning angle, and a suitable cut makes both possible remaining triangles contain such a multiple.
- For nonintegral s, found Shan-Yu's invariant: he starts with an equilateral triangle and always retains a child all of whose angles, in units of θ, are nonintegral. A four-case check proves that every cut has at least one such child.

Current best

Mulan can guarantee victory exactly for $\boxed{\theta = 180°/N}$, where N is an integer with $N \geq 2$.

Full proof

Put

$$s = \frac{180°}{\theta},$$

and measure every angle in units of θ. Thus every triangle has angle sum s, and Mulan's target angle has measure 1.

We first record how a cut changes the angles. Suppose the current triangle is ABC, with normalized angles

$$\angle A = a, \qquad \angle B = b, \qquad \angle C = c,$$

so $a + b + c = s$. If P lies in the interior of BC and $\angle BAP = x$, where $0 < x < a$, then the two triangles have normalized angle triples

$$ABP : (b, x, s - b - x), \qquad ACP : (c, a - x, b + x). \qquad (1)$$

Indeed, the first formula follows from the angle sum, and the two angles at P are supplementary, so the last angle in the second triangle is $b+x$. Conversely, every x with $0 < x < a$ can be realized: the ray from A inside $\angle BAC$ making angle $x\theta$ with AB meets the interior of BC.

1. Sufficiency

Assume that $s = N$ is an integer. Since $0° < \theta < 180°$, we have $N \geq 2$.

First, if a current triangle has an angle equal to m, where m is a positive integer, Mulan can force victory. This is proved by induction on m. For $m = 1$, she has already won. If $m \geq 2$, she cuts from the vertex of that angle so as to split it into angles 1 and $m - 1$. One resulting triangle has angle 1; the other has angle $m - 1$. Hence, whichever triangle Shan-Yu retains, Mulan wins either immediately or by the induction hypothesis.

It remains to show that Mulan can reach this situation from an arbitrary triangle. If one of its normalized angles is already an integer, the preceding paragraph applies. Otherwise, write its angles as a, b, c, all nonintegral, with

$$a + b + c = N.$$

We claim that the vertices can be labeled so that the open interval $(b, a + b)$ contains an integer k.

If some angle, say a, is greater than 1, choose either of the other angles as b. The interval $(b, a+b)$ has length $a > 1$, so it contains an integer. The only remaining possibility is that all three angles are less than 1. Then $N < 3$; since $N \geq 2$ is an integer, $N = 2$. For any choice of two angles a, b, the third angle satisfies $c < 1$, and consequently

$$b < 1 < a + b = 2 - c.$$

Thus $k = 1$ works. This proves the claim.

Choose such a labeling and such an integer k, and set

$$x = k - b.$$

The inequalities $b < k < a + b$ give $0 < x < a$, so the corresponding cut is legal. By (1), one resulting triangle has angle $b + x = k$, while the other has angle

$$N - b - x = N - k.$$

Moreover, $0 < k < N$, since $b > 0$ and $k < a + b = N - c < N$. Therefore k and $N - k$ are positive integers. Whichever triangle Shan-Yu retains, it has a positive integral angle, from which Mulan can force angle 1 by the induction above. Thus Mulan has a finite winning strategy whenever s is an integer.

2. Necessity

Assume now that s is not an integer. Shan-Yu starts with an equilateral triangle. Each of its normalized angles is $s/3$, which is nonintegral: if $s/3$ were an integer, then s would be an integer. In particular, none of these angles is the target value 1.

Shan-Yu maintains the following invariant:

> All three normalized angles of the retained triangle are nonintegral.

Suppose the current normalized angles a, b, c are all nonintegral, and Mulan makes an arbitrary cut. Relabeling the vertices if necessary, its two resulting angle triples have the form in (1):

$$(b, x, s - b - x), \qquad (c, a - x, b + x).$$

We prove that at least one triple consists entirely of nonintegers. Suppose, to the contrary, that each triple contains an integer. Since b and c are nonintegral, we must have

$$x \in \mathbb{Z} \quad \text{or} \quad s - b - x \in \mathbb{Z}, \qquad (2)$$

and

$$a - x \in \mathbb{Z} \quad \text{or} \quad b + x \in \mathbb{Z}. \qquad (3)$$

There are four possible pairings of these alternatives:

- if x and $a - x$ are integers, then a is an integer;
- if x and $b + x$ are integers, then b is an integer;
- if $s - b - x$ and $a - x$ are integers, then their difference $s - a - b = c$ is an integer;
- if $s - b - x$ and $b + x$ are integers, then their sum s is an integer.

Every case contradicts the assumptions. Hence at least one child has three nonintegral normalized angles, and Shan-Yu retains that child.

The invariant can therefore be maintained after every cut. Since the target angle has normalized measure 1, an integer, it never occurs. Thus Mulan cannot guarantee victory when s is nonintegral.

Combining the two directions, Mulan can guarantee victory exactly when

$$\frac{180^\circ}{\theta} = N \in \mathbb{Z},$$

that is,

$$\boxed{\theta = \frac{180^\circ}{N} \quad \text{for an integer } N \geq 2.}$$

IMO 2026 — Problem 5

Solution by GPT-5.6 Sol Pro (xhigh reasoning)

2026-07-22

Source: `results/gpt-5.6-sol-pro/problem-05/current.md` — autonomous run, no human intervention; solution as written by the model.

Problem

Let $\mathbb{R}_{>0}$ be the set of positive real numbers. Determine all functions $f : \mathbb{R}_{>0} \to \mathbb{R}_{>0}$ such that $\sqrt{\frac{x^2+f(y)^2}{2}} \geq \frac{f(x)+y}{2} \geq \sqrt{xf(y)}$ for every $x, y \in \mathbb{R}_{>0}$.

Solution document

Status

solved

Problem

Let $\mathbb{R}_{>0}$ be the set of positive real numbers. Determine all functions $f : \mathbb{R}_{>0} \to \mathbb{R}_{>0}$ such that $\sqrt{\frac{x^2+f(y)^2}{2}} \geq \frac{f(x)+y}{2} \geq \sqrt{xf(y)}$ for every $x, y \in \mathbb{R}_{>0}$.

Approaches tried

- Tested affine and multiplicative candidates. This identified the family $f(x) = x + c$ with $c \geq 0$; direct substitution turns both squared inequality gaps into $(x - y - c)^2$.
- Used the equality-producing substitution $x = f(y)$. It gives the exact second-iterate relation $f(f(y)) = 2f(y) - y$.
- Iterated the second-iterate relation. Every forward orbit is the arithmetic progression $f^n(x) = x + n(f(x) - x)$, so positivity of the codomain forces $f(x) \geq x$.
- Rewrote the upper inequality in terms of the displacement $d(x) = f(x) - x$. For two strictly positive displacements, synchronized their arithmetic orbits to remain a bounded distance apart while tending to infinity. Applying the upper inequality in both orders forces the two displacements to be equal.
- Ruled out coexistence of zero displacement and a common positive displacement. The upper inequality separates points of the two types by more than the positive displacement, while a finite chain in an interval with shorter steps must contain two adjacent points of different types.

- Also examined pointwise bounds obtained from $x = y$ and the lower inequality; these give rough growth restrictions but do not by themselves compare displacements at different points, so they were not used in the final proof.

Current best

All solutions are exactly $\boxed{f(x) = x + c \quad (x > 0)}$, where c is an arbitrary real constant with $c \geq 0$.

Full proof

Define the displacement

$$d(t) := f(t) - t \qquad (t > 0).$$

We first obtain an exact relation. In the given chain put $x = f(y)$. Its leftmost term is then

$$\sqrt{\frac{f(y)^2 + f(y)^2}{2}} = f(y),$$

and its rightmost term is

$$\sqrt{f(y)f(y)} = f(y).$$

Consequently the middle term is equal to $f(y)$, and hence

$$f(f(y)) = 2f(y) - y. \qquad (1)$$

It follows that

$$d(f(y)) = f(f(y)) - f(y) = f(y) - y = d(y). \qquad (2)$$

By induction, (2) gives, for every integer $n \geq 0$,

$$f^n(y) = y + nd(y). \qquad (3)$$

If $d(y) < 0$, then the right-hand side of (3) is nonpositive for every sufficiently large n, contrary to the fact that every iterate of f lies in $\mathbb{R}_{>0}$. Thus

$$d(y) \geq 0 \qquad \text{for every } y > 0. \qquad (4)$$

We next prove that any two positive values of d are equal. Squaring the first inequality in the given chain is legitimate because both sides are positive. On writing

$$a = d(x), \qquad b = d(y),$$

we obtain

$$2x^2 + 2(y + b)^2 \geq (x + a + y)^2,$$

or equivalently

$$(x - y)^2 + 4yb + 2b^2 \geq 2a(x + y) + a^2. \qquad (5)$$

Let $u, v > 0$ satisfy

$$d(u) = a > 0, \qquad d(v) = b > 0.$$

By (2)–(3),

$$d(u + ma) = a, \qquad d(v + nb) = b \qquad (6)$$

for all integers $m, n \geq 0$. For every sufficiently large integer m, put

$$X_m = u + ma, \qquad n_m = \left\lfloor \frac{X_m - v}{b} \right\rfloor, \qquad Y_m = v + n_m b.$$

Then $n_m \geq 0$ and

$$0 \leq X_m - Y_m < b. \qquad (7)$$

In particular, $X_m, Y_m \to \infty$ and $X_m/Y_m \to 1$. By (6), their displacements are respectively a, b. Apply (5) with $(x, y) = (X_m, Y_m)$ and divide by Y_m:

$$\frac{(X_m - Y_m)^2}{Y_m} + 4b + \frac{2b^2}{Y_m} \geq 2a\left(\frac{X_m}{Y_m} + 1\right) + \frac{a^2}{Y_m}.$$

Letting $m \to \infty$ and using (7) yields $4b \geq 4a$, so $b \geq a$. Applying (5) instead with $(x, y) = (Y_m, X_m)$ and dividing by X_m gives, in the same way, $4a \geq 4b$. Therefore $a = b$.

It follows that either d is identically zero, or there is a number $c > 0$ such that

$$d(t) \in \{0, c\} \qquad (t > 0). \qquad (8)$$

We show that in the latter case the value 0 cannot occur. Suppose that $d(p) = c$ and $d(q) = 0$. Substitution of $x = p$, $y = q$, $a = c$, and $b = 0$ into (5) gives

$$(p - q)^2 \geq 2c(p + q) + c^2 > c^2,$$

so every point with displacement c is at distance greater than c from every point with displacement 0:

$$|p - q| > c. \qquad (9)$$

If both types in (8) occurred, choose one point p of the first type and one point q of the second type. Choose an integer $N > |p - q|/c$ and divide the interval from p to q into the points

$$t_i = p + \frac{i}{N}(q - p), \qquad i = 0, 1, \dots, N.$$

All these points are positive, and consecutive ones are at distance $|p - q|/N < c$. Since t_0 and t_N have different displacement types, some consecutive pair must have different types. This contradicts (9). Hence no mixing is possible. Together with (4), we have proved that for some constant $c \geq 0$,

$$d(t) = c \qquad (t > 0),$$

and therefore

$$f(t) = t + c. \qquad (10)$$

It remains to verify these functions. Let $c \geq 0$ and $f(t) = t + c$. For every $x, y > 0$, the arithmetic mean of the two positive numbers x and $y + c$ is

$$\frac{x + y + c}{2} = \frac{f(x) + y}{2}.$$

The root-mean-square–arithmetic-mean and arithmetic-mean–geometric-mean inequalities give

$$\sqrt{\frac{x^2 + (y + c)^2}{2}} \geq \frac{x + y + c}{2} \geq \sqrt{x(y + c)},$$

which is exactly the required chain. Thus all and only the functions in (10), with $c \geq 0$, are solutions.

IMO 2026 — Problem 6

Solution by GPT-5.6 Sol Pro (xhigh reasoning)

2026-07-22

Source: `results/gpt-5.6-sol-pro/problem-06/current.md` — autonomous run, no human intervention; solution as written by the model.

Problem

Let $a_1, a_2, a_3, \ldots$ be an infinite sequence of positive integers greater than 1. Suppose that for all positive integers n, the number a_{n+1} is the smallest positive integer greater than a_n such that $\gcd(a_{n+1}, a_i) > 1$ for every $i = 1, 2, \ldots, n$. Prove that there exist positive integers T and L such that $a_{n+T} = a_n + L$ for every positive integer n. (Note that $\gcd(x, y)$ denotes the greatest common divisor of positive integers x and y.)

Solution document

Status

solved

Problem

Let $a_1, a_2, a_3, \ldots$ be an infinite sequence of positive integers greater than 1. Suppose that for all positive integers n, the number a_{n+1} is the smallest positive integer greater than a_n such that $\gcd(a_{n+1}, a_i) > 1$ for every $i = 1, 2, \ldots, n$. Prove that there exist positive integers T and L such that $a_{n+T} = a_n + L$ for every positive integer n. (Note that $\gcd(x, y)$ denotes the greatest common divisor of positive integers x and y.)

Approaches tried

- Recast the recurrence as a greedy scan of all integers from a_1 onward. An integer is accepted precisely when it is noncoprime to every previously accepted integer; rejected integers have smaller accepted coprime witnesses.
- Passed to finite sets of prime divisors. Acceptance depends only on the prime support, and the family $\mathcal{F}$ of accepted supports is self-dual: $S \in \mathcal{F}$ exactly when S meets every member of $\mathcal{F}$.
- Considered projection to the finitely many primes dividing a_1. This gave restrictions on support types but did not by itself control auxiliary primes.

- Considered proving eventual periodicity from the descending periodic conditions imposed by finite prefixes. Such conditions need not visibly stabilize, so this route did not resolve the core issue.
- Used the inclusion-minimal accepted supports $\mathcal{C}$ and multiplicative weight $w(S) = \prod_{p \in S} p$. Removing one prime from a minimal support produces a rejected support; its smaller coprime witness yields a strict weight descent to a bounded-size minimal support containing the chosen prime.
- Proved, via the infinite sunflower lemma and self-duality, that there are only finitely many minimal supports of any fixed bounded size. The descent then shows that only finitely many primes occur in all minimal supports, so acceptance is periodic modulo their product.

Current best

The sequence is globally translation-periodic. Let $\mathcal{C}$ be the family of inclusion-minimal prime supports of accepted integers. The proof below shows that $\mathcal{C}$ is finite. If $P = \bigcup_{C \in \mathcal{C}} C$ and $L = \prod_{p \in P} p$, then an integer $m \geq a_1$ is a term of the sequence exactly when its prime support contains some $C \in \mathcal{C}$; hence this status is periodic modulo L. Taking T to be the number of accepted residue classes modulo L gives $a_{n+T} = a_n + L$ for every n.

Full proof

We first reinterpret the construction. Starting with a_1, scan the integers $a_1 + 1, a_1 + 2, \ldots$ in increasing order, accepting an integer if and only if it has greatest common divisor greater than 1 with every integer accepted earlier. The accepted integers, in increasing order, are exactly $a_1, a_2, \ldots$. Indeed, this is precisely the minimality condition in the definition of a_{n+1}.

Consequently:

1. any two accepted integers are noncoprime; and
2. every rejected integer $x > a_1$ has a smaller accepted integer coprime to it.

For a positive integer $x > 1$, write

$$\operatorname{supp}(x) = \{p : p \text{ is prime and } p \mid x\}.$$

We claim that, among integers at least a_1, acceptance depends only on the support. Suppose $a_1 \leq x < y$ and $\operatorname{supp}(x) = \operatorname{supp}(y)$. If x is rejected, it has a smaller accepted coprime witness, which is then also coprime to y; hence y is rejected. If x is accepted, every accepted integer smaller than y is noncoprime to x, because accepted integers are pairwise noncoprime. Having the same support as x, the integer y is therefore noncoprime to every earlier accepted integer, so y is accepted. This proves the claim.

Every nonempty finite set of primes is the support of arbitrarily large integers. We may therefore define $\mathcal{F}$ to be the family of nonempty finite prime sets whose integers of sufficiently large size (equivalently, all such integers at least a_1) are accepted. We have the following self-duality property:

$$S \in \mathcal{F} \quad \Longleftrightarrow \quad S \cap R \neq \emptyset \text{ for every } R \in \mathcal{F}. \qquad (1)$$

For the forward implication, choose accepted integers with supports S and R and use pairwise noncoprimality. Conversely, if S meets every member of $\mathcal{F}$, choose an integer $x \geq a_1$ with support S. Every accepted integer smaller than x has support in $\mathcal{F}$, and thus is noncoprime to x. Hence the greedy scan accepts x, proving $S \in \mathcal{F}$.

Let $\mathcal{C}$ be the collection of inclusion-minimal members of $\mathcal{F}$. Every member of $\mathcal{F}$ contains a member of $\mathcal{C}$, because it is a finite set. Thus $\mathcal{C}$ is an intersecting antichain, and

$$S \in \mathcal{F} \quad \iff \quad S \text{ contains some } C \in \mathcal{C}. \qquad (2)$$

We will prove that $\mathcal{C}$ is finite.

We need first a bounded-size finiteness lemma.

Lemma 1. For every positive integer K, there are only finitely many $C \in \mathcal{C}$ such that $|C| \leq K$.

Proof. Suppose instead that there are infinitely many such members. We use the following elementary infinite sunflower fact: every infinite family of distinct sets of cardinality at most K has an infinite subfamily $C_1, C_2, \ldots$ for which there is a fixed set R satisfying

$$C_i \cap C_j = R \qquad (i \neq j). \qquad (3)$$

For completeness, this fact follows by induction on K. If some element occurs in infinitely many sets, restrict to those sets, delete that element, and apply the induction hypothesis. If every element occurs only finitely many times, one can greedily choose infinitely many pairwise disjoint sets, giving $R = \emptyset$.

Apply the fact to the assumed infinite bounded-size subfamily of $\mathcal{C}$. The petals $C_i \setminus R$ are pairwise disjoint. Fix any $H \in \mathcal{C}$. Since $\mathcal{C}$ is intersecting, H meets every C_i. If H were disjoint from R, the finite set H would have to meet every one of the infinitely many pairwise disjoint petals, which is impossible. Thus every $H \in \mathcal{C}$ meets R. Since $\mathcal{C}$ is nonempty, this also shows that R is nonempty.

Every member of $\mathcal{F}$ contains a member of $\mathcal{C}$, so R meets every member of $\mathcal{F}$. By (1), $R \in \mathcal{F}$. Hence R contains some $H_0 \in \mathcal{C}$. But

$$H_0 \subseteq R \subseteq C_i$$

for every i. Since H_0 and C_i belong to the antichain $\mathcal{C}$, this forces $H_0 = C_i$ for every i, contradicting the distinctness of the C_i. This proves Lemma 1. $\square$

For a finite set S of primes, define

$$w(S) = \prod_{p \in S} p,$$

with $w(\emptyset) = 1$.

Lemma 2. If $C \in \mathcal{C}$ and $p \in C$, then there is a $D \in \mathcal{C}$ such that

$$p \in D \qquad \text{and} \qquad w(D \setminus \{p\}) < a_1. \qquad (4)$$

Proof. We first establish a descent step. Let $E \in \mathcal{C}$, let $p \in E$, and suppose

$$w(E \setminus \{p\}) \geq a_1.$$

Set $S = E \setminus \{p\}$. By the inclusion-minimality of E, we have $S \notin \mathcal{F}$. The squarefree integer $w(S)$ has support S, is at least a_1, and is therefore rejected. In fact it is greater than a_1, since a_1 itself is accepted. Hence there exists an accepted integer $y < w(S)$ coprime to $w(S)$. The support of y lies in $\mathcal{F}$, so it contains some $H \in \mathcal{C}$.

We have $H \cap S = \emptyset$. Since $H, E \in \mathcal{C}$ and $\mathcal{C}$ is intersecting, $H \cap E \neq \emptyset$. As $E = S \cup \{p\}$, it follows that $p \in H$. Moreover,

$$w(H) \leq \prod_{q|y} q \leq y < w(S) = w(E \setminus \{p\}). \qquad (5)$$

This is the required descent step.

Starting from C, repeatedly perform this step as long as the current set E satisfies $w(E \setminus \{p\}) \geq a_1$. Every resulting member of $\mathcal{C}$ contains p, and (5) shows that its weight is strictly smaller than the weight of the preceding member. Since these weights are positive integers, the process must terminate. At termination we obtain a set D satisfying (4). This proves Lemma 2. $\square$

Call a member $D \in \mathcal{C}$ terminal if (4) holds for at least one $p \in D$. Every terminal member has bounded cardinality. Indeed, since all primes are at least 2,

$$2^{|D|-1} \leq w(D \setminus \{p\}) < a_1$$

for the corresponding p. In particular, $|D| \leq a_1$, so Lemma 1 shows that there are only finitely many terminal members.

Let P be the union of all terminal members. It is a finite set of primes. If a prime p occurs in any $C \in \mathcal{C}$, Lemma 2 supplies a terminal $D \in \mathcal{C}$ containing p. Therefore every prime occurring in any member of $\mathcal{C}$ belongs to P. It follows that every member of $\mathcal{C}$ is a subset of the finite set P, and hence $\mathcal{C}$ itself is finite, as desired.

Now set

$$L = \prod_{p \in P} p.$$

The set P is nonempty: $\mathcal{F}$ contains the support of a_1, so $\mathcal{C}$ is nonempty. By (2), an integer $m \geq a_1$ is accepted exactly when $\operatorname{supp}(m)$ contains at least one member of $\mathcal{C}$. For each $p \in P$, we have

$$p \mid m \quad \Longleftrightarrow \quad p \mid m + L.$$

It follows that

$$m \text{ is accepted} \quad \Longleftrightarrow \quad m + L \text{ is accepted} \qquad (m \geq a_1). \qquad (6)$$

Let T be the number of residue classes modulo L whose integers at least a_1 are accepted. This is positive because a_1 is accepted. Every half-open interval $(x, x + L]$ contains exactly one representative of each residue class modulo L, and therefore contains exactly T accepted integers. Taking $x = a_n$ and using (6), the endpoint $a_n + L$ is accepted. It is consequently the T-th accepted integer after a_n, so

$$a_{n+T} = a_n + L$$

for every positive integer n. This completes the proof.